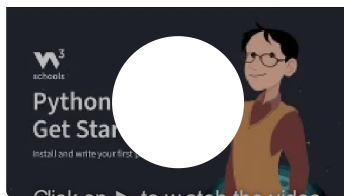


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# HTML URL Encoding Reference

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## URL - Uniform Resource Locator

Web browsers request pages from web servers by using a URL.

The URL is the address of a web page, like: **<https://www.w3schools.com>**.

## URL Encoding (Percent Encoding)

URL encoding converts characters into a format that can be transmitted over the Internet.

URLs can only be sent over the Internet using the ASCII character-set.

Since URLs often contain characters outside the ASCII set, the URL has to be converted into a valid ASCII format.

URL encoding replaces unsafe ASCII characters with a "%" followed by two hexadecimal digits.

URLs cannot contain spaces. URL encoding normally replaces a space with a plus (+) sign or with %20.



to the server. A page at the server will display the received input.

[REMOVE ADS](#)

Try some other input and click Submit again.



## URL Encoding Functions

In JavaScript, PHP, and ASP there are functions that can be used to URL encode a string.

PHP has the `rawurlencode()` function, and ASP has the `Server.URLEncode()` function.

In JavaScript you can use the **`encodeURIComponent()`** function.

Click the "URL Encode" button to see how the JavaScript function encodes the text.

**Note:** The JavaScript function encodes space as `%20`.

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## ASCII Encoding Reference



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| Character<br><a href="#">REMOVE ADS</a> | From Windows-1252 | From UTF-8 |
|-----------------------------------------|-------------------|------------|
| space                                   | %20               | %20        |
| !                                       | %21               | %21        |
| "                                       | %22               | %22        |
| #                                       | %23               | %23        |
| \$                                      | %24               | %24        |
| %                                       | %25               | %25        |
| &                                       | %26               | %26        |
| '                                       | %27               | %27        |
| (                                       | %28               | %28        |
| )                                       | %29               | %29        |
| *                                       | %2A               | %2A        |
| +                                       | %2B               | %2B        |
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# URL Encoding Reference

The ASCII control characters **%00-%1F** were originally designed to control hardware devices.

Control characters have nothing to do inside a URL.

| ASCII Character | Description         | URL-encoding |
|-----------------|---------------------|--------------|
| NUL             | null character      | %00          |
| SOH             | start of header     | %01          |
| STX             | start of text       | %02          |
| ETX             | end of text         | %03          |
| EOT             | end of transmission | %04          |
| ENQ             | enquiry             | %05          |
| ACK             | acknowledge         | %06          |
| BEL             | bell (ring)         | %07          |
| BS              | backspace           | %08          |
| HT              | horizontal tab      | %09          |
| LF              | line feed           | %0A          |
| VT              | vertical tab        | %0B          |
| FF              | form feed           | %0C          |
| CR              | carriage return     | %0D          |
| SO              | shift out           | %0E          |
| SI              | shift in            | %0F          |
| DLE             | data link escape    | %10          |

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|                                   |                        |     |
|-----------------------------------|------------------------|-----|
| DC3<br><a href="#">REMOVE ADS</a> | device control 3       | %13 |
| DC4                               | device control 4       | %14 |
| NAK                               | negative acknowledge   | %15 |
| SYN                               | synchronize            | %16 |
| ETB                               | end transmission block | %17 |
| CAN                               | cancel                 | %18 |
| EM                                | end of medium          | %19 |
| SUB                               | substitute             | %1A |
| ESC                               | escape                 | %1B |
| FS                                | file separator         | %1C |
| GS                                | group separator        | %1D |
| RS                                | record separator       | %1E |
| US                                | unit separator         | %1F |

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